



COMP4121 Advanced Algorithms

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Some Examples

An example of a Maximum Likelihood Estimation

- Assume that we have n sensors such that the errors of readings $X_1, \dots, X_n$ of these n sensors are independent, unbiased and normally distributed, but with different and **known** standard deviations $\sigma_1, \dots, \sigma_n$ (for example, we have tested them all in a lab doing many measurements and comparing their readings with the true value obtained from a precise instrument).
- Assume that, using these sensors, we have obtained measurements $X_1, \dots, X_n$, which we would like to use to estimate the true value of the quantity being measured.
- Let the true value of the measured quantity be equal to μ .
- Since the sensors are unbiased the expected value of the readings of all sensors is equal to μ , i.e., $E[X_i] = \mu$.
- We can now compute the **likelihood** that a particular μ produced measurements $X_1, X_2, \dots, X_n$ and then pick the value of μ for which such a likelihood is the largest.

An example of a Maximum Likelihood Estimation

- Since the errors are independent and normally distributed and since $E[X_i] = \mu$, the probability to obtain readings $\mathbf{X} = X_1, \dots, X_n$ is equal to

$$\mathcal{L}_n(\mu|\mathbf{X}) = \prod_{i=1}^n \frac{1}{\sigma_i \sqrt{2\pi}} e^{-\frac{1}{2} \frac{(X_i - \mu)^2}{\sigma_i^2}} = \left(\prod_{i=1}^n \frac{1}{\sigma_i \sqrt{2\pi}} \right) e^{-\frac{1}{2} \sum_{i=1}^n \frac{(X_i - \mu)^2}{\sigma_i^2}}$$

- Differentiating with respect to μ and setting the derivative equal to zero we get

$$\frac{d}{d\mu} \mathcal{L}_n(\mu|\mathbf{X}) = \left(\prod_{i=1}^n \frac{1}{\sigma_i \sqrt{2\pi}} \right) e^{-\frac{1}{2} \sum_{i=1}^n \frac{(X_i - \mu)^2}{\sigma_i^2}} \sum_{i=1}^n \frac{(X_i - \mu)}{\sigma_i^2}$$

- Thus,

$$\frac{d}{d\mu} \mathcal{L}_n(\mu|\mathbf{X}) = 0 \quad \Leftrightarrow \quad \sum_{i=1}^n \frac{X_i}{\sigma_i^2} - \mu \sum_{i=1}^n \frac{1}{\sigma_i^2} = 0 \quad \Leftrightarrow \quad \mu = \frac{\sum_{i=1}^n \frac{X_i}{\sigma_i^2}}{\sum_{j=1}^n \frac{1}{\sigma_j^2}}$$

An example of a Maximum Likelihood Estimation

- Since the Maximum Likelihood estimate of the true value is

$$\mu = \frac{\sum_{i=1}^n \frac{X_i}{\sigma_i^2}}{\sum_{i=1}^n \frac{1}{\sigma_i^2}} = \sum_{i=1}^n \frac{\frac{1}{\sigma_i^2}}{\sum_{j=1}^n \frac{1}{\sigma_j^2}} X_i$$

we see that the maximum likelihood estimate is a weighted mean of the readings of **all** sensors where each sensor's reading is weighted inversely proportionally to its variance.

- Let us find the variance of such an estimator:

$$\begin{aligned} V(M(\mathbf{X})) &= E \left(\sum_{i=1}^n \frac{\frac{1}{\sigma_i^2}}{\sum_{j=1}^n \frac{1}{\sigma_j^2}} X_i - \mu \right)^2 = E \left(\sum_{i=1}^n \frac{\frac{1}{\sigma_i^2} (X_i - \mu)}{\sum_{k=1}^n \frac{1}{\sigma_k^2}} \right)^2 \\ &= E \left(\sum_{i,j=1}^n \frac{\frac{1}{\sigma_i^2} (X_i - \mu)}{\sum_{k=1}^n \frac{1}{\sigma_k^2}} \frac{\frac{1}{\sigma_j^2} (X_j - \mu)}{\sum_{k=1}^n \frac{1}{\sigma_k^2}} \right) \end{aligned}$$

An example of a Maximum Likelihood Estimation

- Since the errors of X_i' s are pairwise independent, we have $E((X_i - \mu)(X_j - \mu)) = 0$ for $i \neq j$; thus, from

$$V(M(\mathbf{X})) = E \left(\sum_{i,j=1}^n \frac{\frac{1}{\sigma_i^2}(X_i - \mu)}{\sum_{k=1}^n \frac{1}{\sigma_k^2}} \frac{\frac{1}{\sigma_j^2}(X_j - \mu)}{\sum_{k=1}^n \frac{1}{\sigma_k^2}} \right)$$

we obtain

$$V(M(\mathbf{X})) = E \left(\sum_{i=1}^n \frac{\frac{1}{\sigma_i^4}(X_i - \mu)^2}{\left(\sum_{j=1}^n \frac{1}{\sigma_j^2} \right)^2} \right) = \sum_{i=1}^n \frac{\frac{1}{\sigma_i^4} \sigma_i^2}{\left(\sum_{j=1}^n \frac{1}{\sigma_j^2} \right)^2} = \frac{1}{\sum_{j=1}^n \frac{1}{\sigma_j^2}}.$$

An example of a Maximum Likelihood Estimation

- Since for all i

$$V(M(\mathbf{X})) = \frac{1}{\sum_{j=1}^n \frac{1}{\sigma_j^2}} < \frac{1}{\frac{1}{\sigma_i^2}} = \sigma_i^2$$

we get that also $V(M(\mathbf{X})) < \min_{1 \leq i \leq n} \sigma_i^2$.

- Thus, we see that see that our ML estimate is more accurate than the best of our sensors!
- Can we find a more accurate estimator which is also unbiased?
- It turns out that in this case the ML estimator achieves the smallest possible variance among all unbiased estimators.
- Such a lower bound for the variances of all unbiased estimators is given by the Cramer - Rao Theorem which you can look up in the lecture notes on Iterative Filtering which you can find at the class website, look at page 6 of https://www.cse.unsw.edu.au/~cs4121/lectures_2019/IF.pdf

Application of Markov's Inequality: Chebyshev's Inequality

(Markov was a student of Chebyshev)

- **Theorem:** Let X be any random variable with a finite expectation $E[X]$ and a finite variance $V[X]$. Then

$$\mathcal{P}(|X - E[X]| \geq t) \leq \frac{V[X]}{t^2}$$

- **Proof:** Note that

$$\mathcal{P}(|X - E[X]| \geq t) = \mathcal{P}((X - E[X])^2 \geq t^2)$$

- Also, $Y = (X - E[X])^2$ is a non-negative random variable with a finite expectation $E[Y] = E((X - E[X])^2) = V[X]$.
- Thus, we can apply the Markov inequality to Y to obtain

$$\mathcal{P}(|X - E[X]| \geq t) = \mathcal{P}((X - E[X])^2 \geq t^2) = \mathcal{P}(Y \geq t^2) \leq \frac{E[Y]}{t^2} = \frac{V[X]}{t^2}$$

Law of Large Numbers

- If we get a large number of independent samples $X_1, \dots, X_n$ of a random variable X with an expectation $E[X]$ we would expect that the mean of these samples should be close to the expected value $E[X]$ of X . The Law of large numbers bounds the probability that such a mean is further away from $E[X]$ than a prescribed value ε :

$$\mathcal{P} \left(\left| \frac{X_1 + X_2 + \dots + X_n}{n} - E[X] \right| > \varepsilon \right) \leq \frac{V[X]}{n\varepsilon^2}$$

- **Proof:** By the Chebyshev inequality,

$$\mathcal{P} \left(\left| \frac{X_1 + \dots + X_n}{n} - E \left[\frac{X_1 + X_2 + \dots + X_n}{n} \right] \right| > \varepsilon \right) \leq \frac{V \left[\frac{X_1 + X_2 + \dots + X_n}{n} \right]}{\varepsilon^2}$$

- We now use the fact that the expectation is a linear operator and all X_i are equally distributed to conclude that

$$E \left[\frac{X_1 + X_2 + \dots + X_n}{n} \right] = \frac{E[X_1] + E[X_2] + \dots + E[X_n]}{n} = \frac{n E[X]}{n} = E[X]$$

- On the other hand, variance of any random variable X satisfies $V[aX] = a^2V[X]$.
- Also, for INDEPENDENT variables $X_1, \dots, X_n$ we have $V[X_1 + \dots + X_n] = V[X_1] + \dots + V[X_n]$.

Law of Large Numbers

- Thus

$$\mathcal{P} \left(\left| \frac{X_1 + \dots + X_n}{n} - E \left[\frac{X_1 + \dots + X_n}{n} \right] \right| > \varepsilon \right) \leq \frac{V \left[\frac{X_1 + \dots + X_n}{n} \right]}{\varepsilon^2}$$

implies

$$\mathcal{P} \left(\left| \frac{X_1 + \dots + X_n}{n} - E[X] \right| > \varepsilon \right) \leq \frac{\frac{n V[X]}{n^2}}{\varepsilon^2} = \frac{V[X]}{n \varepsilon^2}$$

- To summarise, if $X_1, \dots, X_n$ are independent, equally distributed random variables with a finite expectation $E[X_i] = \mu$ and variance $V[X_i] = v$, then

$$E \left[\frac{X_1 + \dots + X_n}{n} \right] = \mu;$$

$$V \left[\frac{X_1 + \dots + X_n}{n} \right] = \frac{v}{n}$$

$$\mathcal{P} \left(\left| \frac{X_1 + \dots + X_n}{n} - \mu \right| > \varepsilon \right) \leq \frac{v}{n \varepsilon^2}$$